

Forecasting in Healthcare - Training Plan (3 Phases)

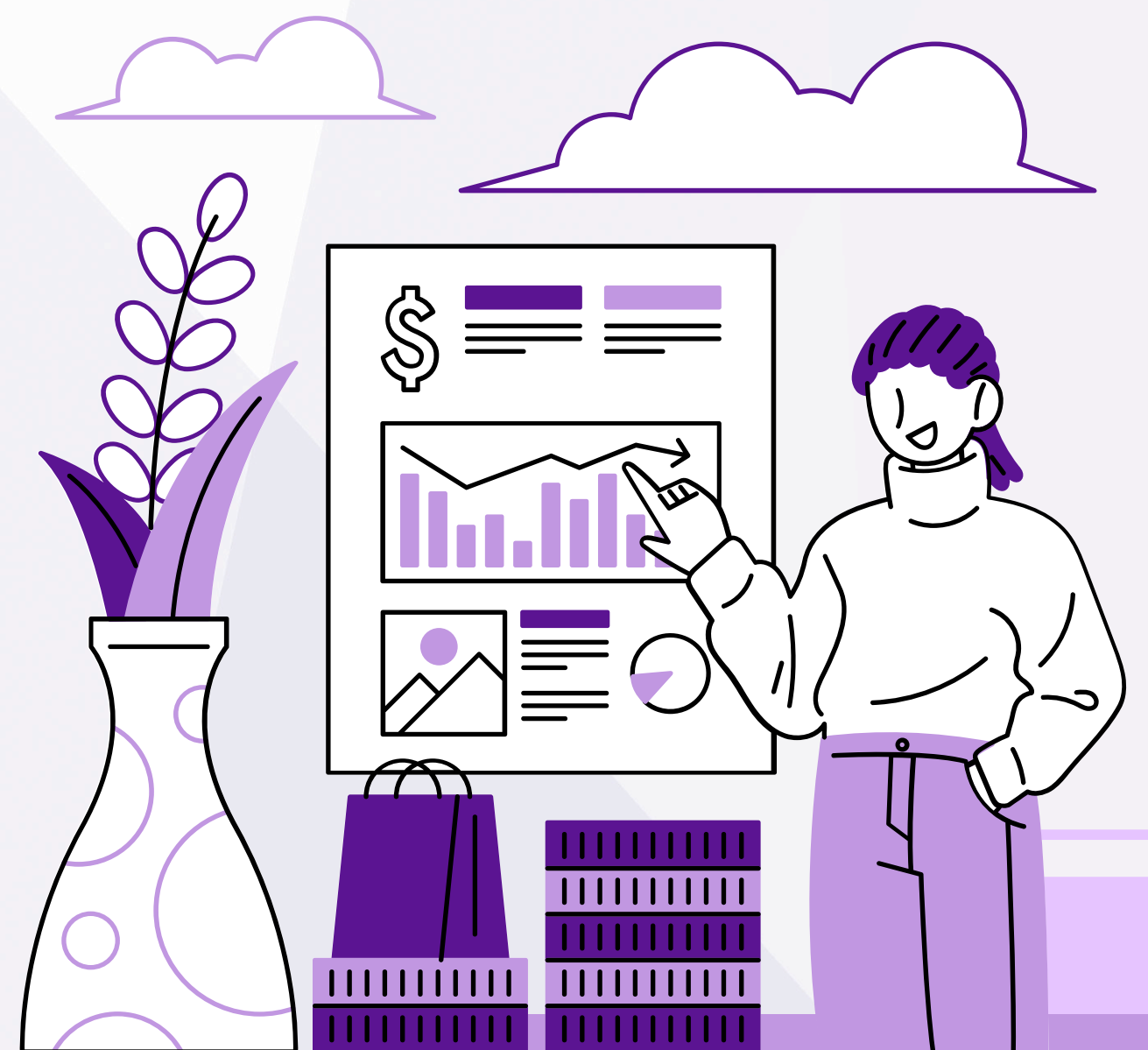
PHASE 1

- **Pharma Industry Overview**

1. Pharma value chain
2. Clinical development pathwa
3. Clinical Development Spend vs Return
4. Market Access & Pricing Concepts
5. 3 P's & Patient Journey
6. Treatment/Management & Therapy Area Overview
7. Treatment Guidelines

- **Basics Statistics and Forecasting**

8. Correlation?
9. Regression techniques
10. Time horizons in forecasting
11. Forecasting Cycle (Define|Model|Analyze|Present)
12. Forecasting approaches (pipeline and inline forecasting)
13. Treatment Algorithm
14. Epidemiology-based forecast
15. Concept of Patient Tower
16. Patient-based forecast
17. Sales-based forecast
18. Time Series Analysis
19. Time series decomposition (Trend, Seasonality, Cyclicity, Irregularities)
20. Moving Average model
21. Forecasting Performance Measures (MAPE, MAD, RMSE)
22. Exponential Smoothing
23. Holt Winter's Model
24. ARMA/ARIMA
- **Advanced Excel**
25. Vlookup, Hlookups
26. Index, match
27. Offset
28. IF Statements Sum, nested IF etc.
29. Indirect/Choose
30. Subtotals and Grouping
31. Pivot Tables
32. Chart Techniques



PHASE 2

- **Applications of Forecasting in Pharma**

1. Basic Terminologies (Patient Filtering) - Prevalence, Incidence, Diagnosis, Treatment, Persistence, Compliance, DoT, Concomitance, Polypharmacy
2. Forecast Assumptions - Secondary & Primary Research (publications/ literature; analogues; analyst reports; syndicated reports; PMR)
3. Product Archetype & Conjoint Analysis
4. Current market trends
5. Drivers of market share
6. Impact of Promotional spending on market share
7. Launch Order/Order of Entry
8. Regulatory approval date
9. Peak Sales & Year to Peak
10. Cannibalization
11. Concept of uptake
12. Therapy Class uptake (S-curve) & Brand Market Share uptake (R-curve)
13. Compliance and persistence
14. Flow/State models (Concepts of new, repeat, switch etc.)
15. Concept of Erosion: Biosimilar & generics impact
16. Patients & Physicians and Payer Impact
17. Risk & Uncertainty

- **Story Boarding**

18. Presentation Skills/Tell your forecasting story
19. Judgement Versus Modelling
20. Trend-based extrapolation versus Causal/event-based
21. Visualizations

- **Forecast Generation Process**

22. Forecast flow (model/structure)
23. Forecast assumption validation
24. Decision analysis framework
25. Sensitivities & Tornado Chart, Waterfall Diagram
26. Scenario Simulations



PHASE 3

- **Visualization**

1. Dashboarding
2. Visualization using Tableau, Power BI, Qlik etc.

- **PowerPoint**

3. Basic Presentation Techniques
4. Proper use of Fonts and Colour Schemes
5. Charts
6. Tornado Charts
7. Combination charts
8. Waterfalls
9. Bubble charts
10. Box Charts
11. Animation

- **Latest Trends in Drug Forecasting**

12. Predictive Analytics
13. AI, ML
14. Using tools such as R or Python
15. Decision Trees, Random Forest etc.

- **Workshop 2-3 Weeks Later**

